A

PROJECT REPORT

ON

"Pick & Place Robotic Arm Vehicle"

SUBMITTED BY

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UNDER THE GUIDANCE OF

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DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING



Shanti Education Society's

A.G. PATIL POLYTECHNIC INSTITUTE, SOLAPUR.

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2023-24

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CERTIFICATE

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"Pick & Place Robotic Arm Vehicle"

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ABSTRACT

The rapid advancement of automation technologies has led to increased demand for efficient and versatile robotic solutions in various industries.This project focuses on the design and implementation of a pick-and-place robotic arm vehicle aimed at enhancing productivity and flexibility in industrial settings.The robotic arm vehicle is equipped with a multi-degree-of-freedom manipulator capable of picking up objects of different shapes, sizes, and weights, and precisely placing them at predefined locations.The vehicle incorporates sensors for environment perception,enabling it to navigate safely in dynamic environments while avoiding obstacles and collisions.Key components of the project include the mechanical design of the robotic arm and vehicle chassis, selection and integration of actuators and sensors, development of control algorithms for motion planning and object manipulation, and integration of a user-friendly interface for operation and monitoring. The proposed system offers several advantages,including increased efficiency, reduced labour costs, improved workplace safety, and enhanced adaptability to changing production requirements. The project contributes to the ongoing advancement of robotic technologies and demonstrates the feasibility and practicality of integrating robotic arm vehicles into industrial workflows.

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1. Introduction

In the rapidly evolving landscape of industrial automation, efficiency and precision are paramount. Introducing the Pick and Place Robotic Arm Vehicle, a groundbreaking innovation poised to redefine the way materials are handled and processes are streamlined. Equipped with state-of-the-art robotic technology, this vehicle seamlessly integrates mobility with robotic manipulation, offering unparalleled flexibility and adaptability in various industrial settings. Whether in manufacturing plants, warehouses, or logistics centres, this versatile solution promises to optimize workflows and maximize productivity. With its advanced sensing capabilities and intelligent algorithms, the Pick and Place Robotic Arm Vehicle excels in identifying, picking, and placing objects with utmost accuracy and speed. Its robust design ensures reliability even in demanding environments, while its intuitive interface simplifies operation and maintenance.

Say goodbye to manual labour and hello to a new era of automation with the Pick and Place Robotic Arm Vehicle. Join us in revolutionizing industries, one precise movement at a time.

A pick-and-place robotic arm is a mechanical system designed to perform the task of picking up objects from one location and placing them in another. It consists of multiple segments connected, similar to a human arm, and is equipped with motors, sensors, and grippers.

The robotic arm is programmed to move in a precise and controlled manner. It can be guided by various input methods, such as a computer interface or remote control. The arm uses its grippers to grasp objects securely, and then it can move them to a different location.

Pick-and-place robotic arms are commonly used in industries such as manufacturing, logistics, and assembly lines. They automate repetitive tasks that involve moving objects, saving time and reducing the risk of human error. These robotic arms can handle a wide range of objects, from small components to larger items, with accuracy and efficiency.

2. Block Diagram

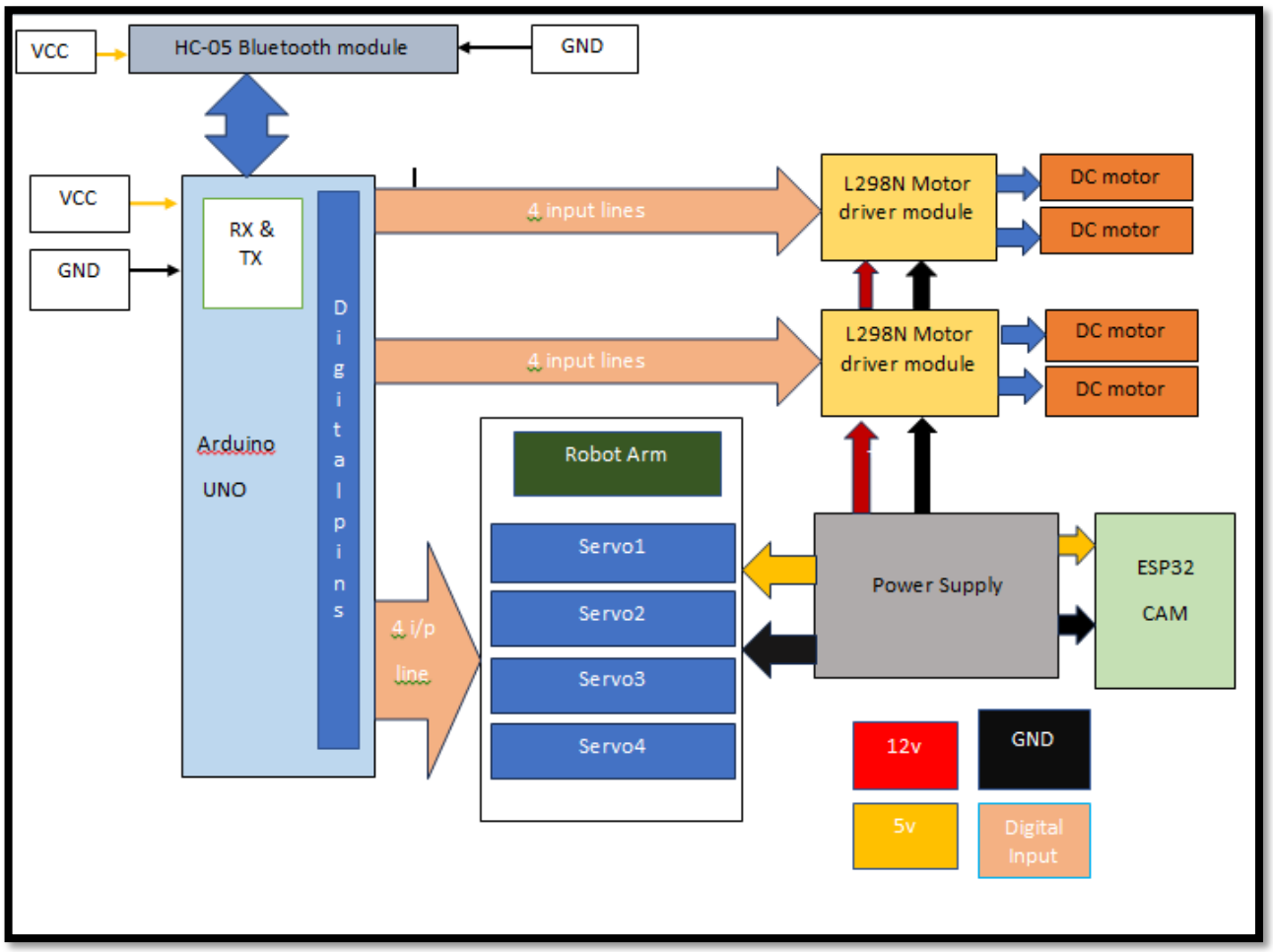


Fig. No. 1. Pick and Place Robotic Arm Vehicle.

2.1 Components Used.

The Robot consists of the following Components:

- Arduino UNO board
- Servo motors (SG90 & MG996R)
- Bluetooth module
- 775 DC motors
- Lithium-ion Battery
- L298N motor driver
- Switches
- Mecanum wheels
- 3D printed robot arm

2.2 Working:

A pick and place robotic arm vehicle using Arduino typically works by utilizing servo motors or stepper motors to control the movement of the robotic arm. The Arduino board serves as the brain, receiving input from sensors such as proximity sensors or cameras to detect objects and determine their positions. The Arduino then processes this information and sends commands to the motors to move the robotic arm accordingly. Once the arm reaches the desired position, it uses a gripper mechanism to pick up the object. After picking up the object, the arm can then move to another location and release the object using the gripper mechanism. Overall, the Arduino controls the entire operation of the robotic arm vehicle, from detecting objects to picking them up and placing them elsewhere, based on the programmed instructions.

A pick and place robotic arm vehicle typically operate using a combination of sensors, actuators, and control systems. Here's a basic overview of how it works:

Sensing: The vehicle uses various sensors such as cameras, LiDAR, or ultrasonic sensors to perceive its environment. These sensors help the robot detect objects, obstacles, and the precise location of items to be picked up and placed.

Planning: Based on the data gathered from sensors, the robot plans its path to navigate the environment efficiently. This involves algorithms to determine the best route to reach the target object and avoid obstacles.

Manipulation: Once the robot reaches the target object, its robotic arm, equipped with grippers or suction cups, picks up the object securely. The arm may have multiple degrees of freedom to maneuver in different directions.

Transportation: After grasping the object, the robot transports it to the desired location. This may involve precise movements to ensure the object is placed accurately.

Release: Once the robot reaches the destination, it releases the object using its gripper or suction cups.

Repeat: The robot continues this process, picking up and placing objects as instructed, until the task is complete.

The entire process is controlled by software that coordinates the actions of the sensors, actuators, and the robotic arm to perform the desired pick and place tasks efficiently and accurately.

3. Circuit Diagram

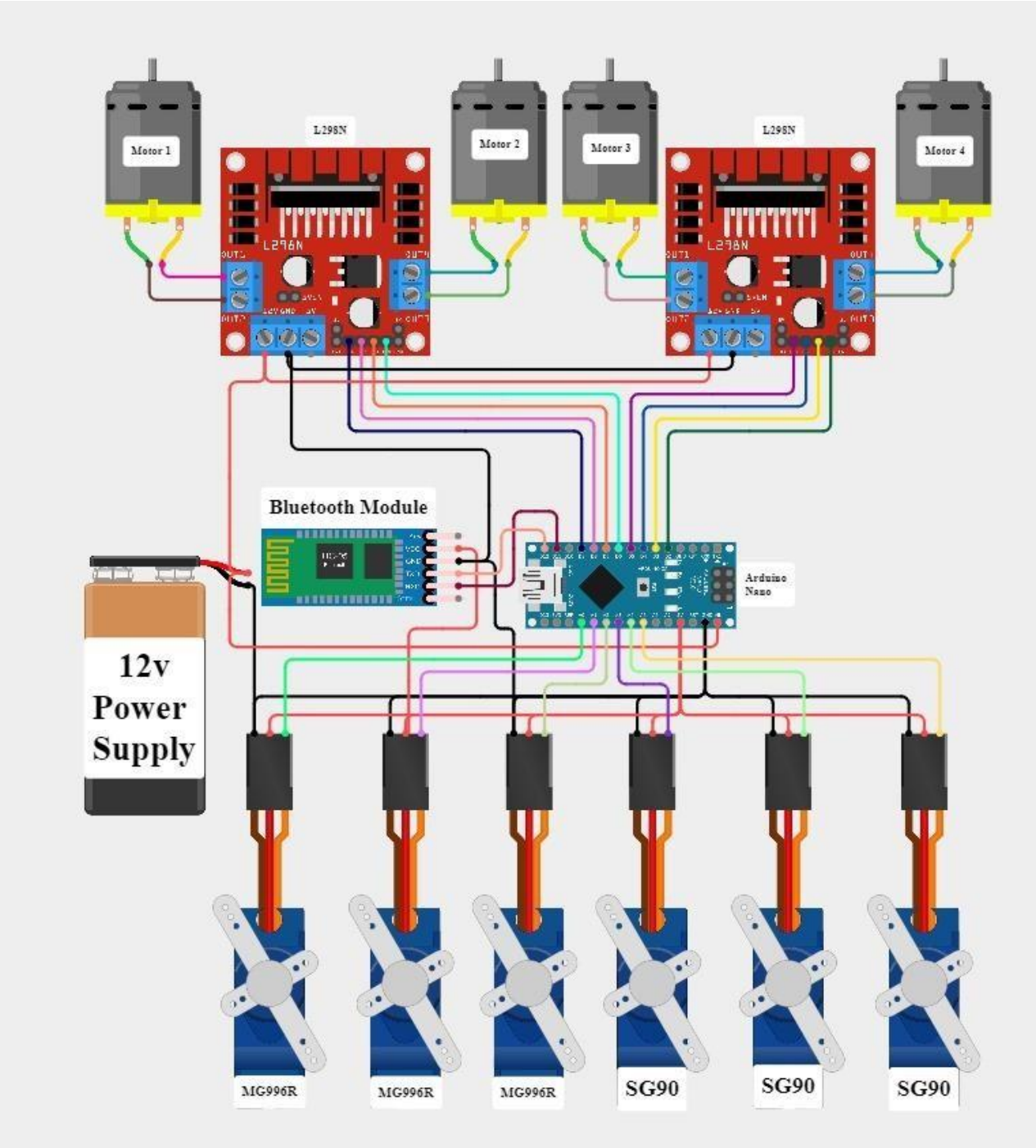


Fig. No. 2. Circuit Diagram

4. Hardware Specification

4.1 Arduino Nano

The Arduino Nano is a small, complete, and breadboard-friendly board based on the ATmega328P released in 2008. It offers the same connectivity and specs of the Arduino Uno board in a smaller form factor.

The Arduino Nano is equipped with 30 male I/O headers, in a DIP-30-like configuration, which can be programmed using the Arduino Software integrated development environment (IDE), which is common to all Arduino boards and running both online and offline. The board can be powered through a type-B mini-USB cable or from a 5 V battery.

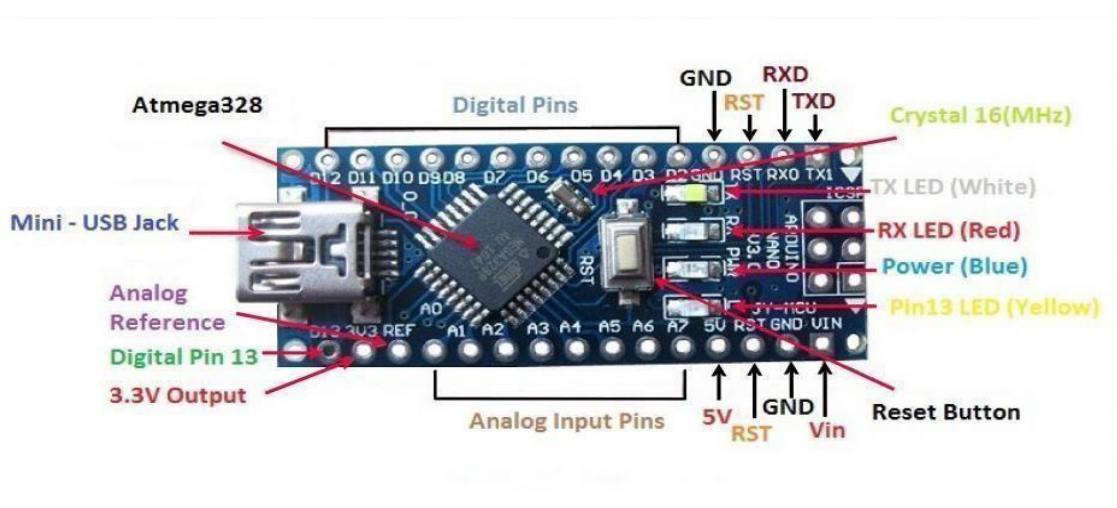


Fig. No. 3. Arduino Nano

- **Specification of Arduino Nano**

Parameters	Value
MICROCONTROLLER	ATmega328
ARCHITECTURE	AVR
OPERATING VOLTAGE RANGE	1.8V to 5.5 V
FLASH MEMORY	32 KB of which 2 KB used by bootloader
SRAM	2 KB

CLOCK SPEED	16 MHz
ANALOG IN PINS	8
EEPROM	1 KB
DC CURRENT PER I/O PINS	40 mA (I/O Pins)
INPUT VOLTAGE	7-12V
DIGITAL I/O PINS	22 (6 of which are PWM)
PWM OUTPUT	6

- **Pin Diagram of ATmega328**

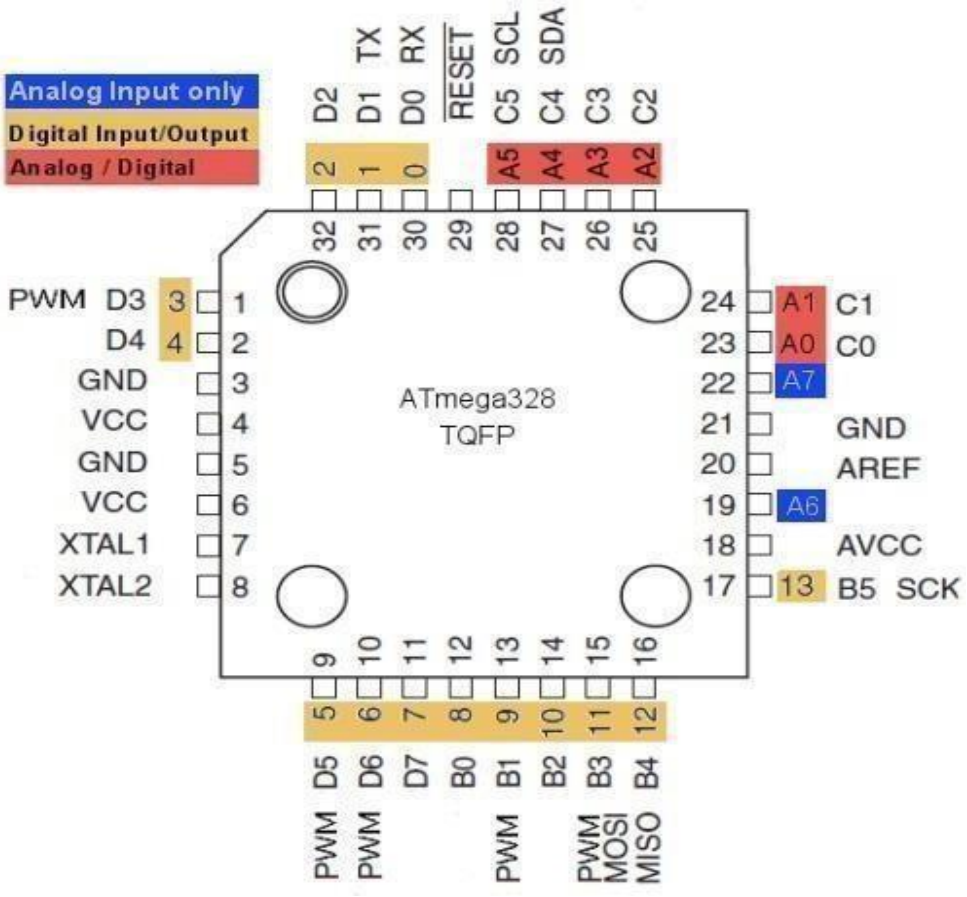


Fig. no. 4. Pin Diagram of ATmega328

4.2 Servo Motor

There are several types of motors as steppers, or stepper motors, and the servomotors. Within the latter there are some really interesting models, such as the case of Servo SG90. A servo that can be ideal for the first projects, practices with this type of device, learning, simple robot control, etc. Also, its power requirements are quite low, it can even be powered from a Adrien plate or from a PC USB port to 5v.

The SG90 servo is a miniature servo, with some very compact dimensions to be able to integrate in projects where space is important. In addition, it is economical and easy to use, with a very low energy demand, so it is also possible to use it in embedded, IoT or other low consumption applications.

As for the Servo SG90, this servo motor includes a universal type S connector that will be able to fit in most of the commercial devices. It is made up of 3 wires with colors that identify what each one is used for:



Fig. No. 5. Servo Motor

➢ Specifications

- Weight: 9 g
- Dimension: 22.2 x 11.8 x 31 mm approx.
- Stall torque: 1.8 kgf·cm
- Operating speed: 0.1 s/60 degree
- Operating voltage: 4.8 V (~5V)
- Dead band width: 10 μs
- Temperature range: 0 ºC – 55 ºC Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is all the way to the left. ms pulse) is all the way to the right, ""-90" (~1ms pulse) is all the way to the left.

The TP SG90 is similar in size and weight to the Hitec HS-55, and is a good choice for most park flyers and helicopters. Hobbyists from around the world has used the SG90 on famous planes like GWS Slow Stick, E-Flite Airplanes, Great Planes, Thunder Tiger, Align, EDF jets and more. If you are looking for a servo that won't break your arm or leg, this is the perfect choice for you! The TP SG90 servo weighs 0.32 ounces (9.0 grams). Total weight with wire and connector is 0.37 ounces (10.6 grams). The TP SG90 has the universal "S" type connector that fits most receivers, including Futaba, JR, GWS, Cirrus, Blue Bird, Blue Arrow, Corona, Berg and Hitec. The wire colours are Red = Battery (+) Brown = Battery (-) Orange = Signal

➢ Basic Information

- Modulation: Analog
- Torque: 4.8V: 25.0 oz-in (1.80 kg-cm)
- Speed: 4.8V: 0.10 sec/60°
- Weight: 0.32 oz (9.0 g)
- Dimensions: Length: 0.91 in (23.1 mm)
- Width: 0.48 in (12.2 mm)
- Height:1.14 in (29.0 mm)
- Motor Type: 3-pole
- Gear Type: Plastic
- Rotation/Support: Bushing

➢ 9g Tower Pro

Servo Tower Pro 9g servo These Micro strong and made to last, great for planes. The servos include 3 servo arms.

- Dimension: 26mm*13mm*24mm
- Weight: 9G
- Operating Speed: 0.12sec/60degree(4.8V);0.11sec/60degree(6V)
- Stall Torque: 1.2kg/cm or 17oz-in. (4.8V) 1.6kg/cm or 22oz-in.(6.0V)
- Operating Voltage: 4.8V~6.0V
- Features: 3 pole wire, all nylon gear, connector wire length: 15cm

4.3 775 DC Motor



Fig.6. 775 DC Motor

➢ Description-

The RS775 DC Motor, a high-torque motor designed to deliver exceptional power and precision. Operating within a voltage range of 12V to 24V, this motor offers a torque range of 1800RPM to 3200RPM, providing you with the force and strength needed to tackle demanding applications. Get ready to elevate your projects to new heights with the RS775 DC Motor and experience the power of high-torque performance.

➢ Specifications-

- Voltage: 12v
- Speed: 1500-35000 RPM
- Output Power: 40-200W
- Dimension: 45 dia x 60 mm length

4.4 Bluetooth Module HC 05

The HC-05 is a class 2 Bluetooth module designed for transparent wireless serial communication. It is pre-configured as a slave Bluetooth device. Once it is paired to a master Bluetooth device such as PC, smart phones and tablet, its operation becomes transparent to the user. All data received through the serial input is immediately transmitted over the air. When the module receives wireless data, it is sent out through the serial interface exactly at it is received. No user code specific to the Bluetooth module is needed at all in the user microcontroller program.

The HC-05 supports two work modes: Command and Data mode. The work mode of the HC-05 can be switched by the onboard push button. The HC-05 is put in Command mode if the push button is activated. In Command mode, user can change the system parameters (e.g. pin code, baud rate, etc) using host controller itself of a PC running terminal software using a serial to TTL converter. Any changes made to system parameters will be retained even after power is removed. Power cycle the HC-05 will set it back to Data Mode. Transparent UART data transfer with a connected remote device occurs only while in Data Mode.

The HC-05 can be re-configured by the user to work as a master Bluetooth device using a set of AT commands. Once configured as master, it can automatically pair with a HC05 in its default slave configuration or a HC-06 module, allowing an point to point serial communications.

The HC-05 will work with supply voltage of 3.6VDC to 6VDC, however, the logic level of RXD pin is 3.3V and is not 5V tolerant. A Logic Level Converter is recommended to protect the sensor if connect it to a 5V device (e.g Arduino Uno and Mega). The power to the HC-05 will cut off if the "EN" pin is pulled to logic 0.

It is used for many applications like wireless headset, game controllers, wireless mouse, wireless keyboard, and many more consumer applications. It has range up to <100m which depends upon transmitter and receiver, atmosphere, geographic & urban conditions. It is IEEE 802.15.1 standardized protocol, through which one can build wireless Personal Area Network (PAN). It uses frequency-hopping spread spectrum (FHSS) radio technology to send data over air.It uses serial communication to communicate with devices. It communicates with microcontroller using serial port (USART).

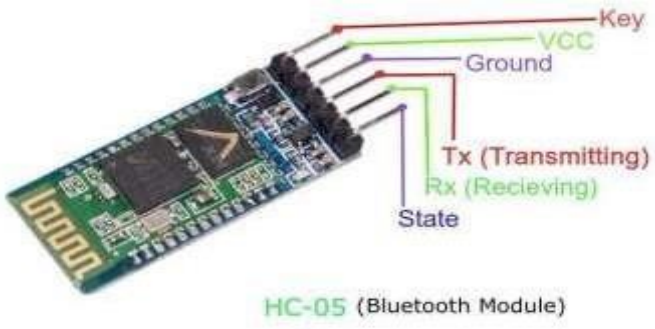


Fig. No. 7. Bluetooth Module HC05

Bluetooth serial modules allow all serial enabled devices to communicate with each other using Bluetooth.

It has 6 pins,

➢ Pin-out Configuration

1. **Key/EN:** It is used to bring Bluetooth module in AT commands mode. If Key/EN pin is set to high, then this module will work in command mode. Otherwise by default it is in data mode. The default baud rate of HC-05 in command mode is 38400bps and 9600 in data mode.

HC-05 module has two modes,

1. **Data mode:** Exchange of data between devices.
2. **Command mode:** It uses AT commands which are used to change setting of HC-05. To send these commands to module serial (USART) port is used.
2. **VCC:** Connect 5 V or 3.3 V to this Pin.
3. **GND:** Ground Pin of module.
4. **TXD:** Transmit Serial data (wirelessly received data by Bluetooth module transmitted out serially on TXD pin)
5. **RXD:** Receive data serially (received data will be transmitted wirelessly by Bluetooth module).
6. **State:** It tells whether module is connected or not.

➢ HC-05 module Information

- HC-05 has **red LED** which indicates **connection status**, whether the Bluetooth is connected or not. Before connecting to HC-05 module this red LED blinks continuously in a periodic manner. When it gets connected to any other Bluetooth device, its blinking slows down to two seconds.
- This module **works on 3.3V**. We can connect 5V supply voltage as well since the module has on board 5 to 3.3 V regulator.
- As HC-05 Bluetooth module has **3.3V level for RX/TX** and microcontroller can detect 3.3 V level, so, no need to shift transmit level of HC-05 module. But we need to shift the transmit voltage level from microcontroller to RX of HC-05 module.
- The data transfer rate of HC-05 module can vary up to **1Mbps** is in the **range of 10 meters**.

➢ Specifications

- Bluetooth version: 2.0 + EDR (Enhanced Data Rate)
- Frequency: 2.4 GHz ISM band
- Modulation: GFSK (Gaussian Frequency Shift Keying)
- Transmit power: Class 2 (up to 4 dBm)
- Sensitivity: -80 dBm typical
- Range: approximately 10 meters (or 33 feet) in open air
- Profiles supported: SPP (Serial Port Profile), HID (Human Interface Device) and others
- Operating voltage: 3.3V to 5V DC
- Operating current: less than 50mA
- Standby current: less than 2.5mA
- Sleep current: less than 1mA
- Interface: UART (Universal Asynchronous Receiver/Transmitter)
- Baud rates: 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200, 230400, and 460800
- Operating temperature: -20°C to 75°C (-4°F to 167°F)

4.5 Lithium Ion-Battery

A lithium-ion or Li-ion battery is a type of rechargeable battery which uses thereversible reduction of lithium ions to store energy. The anode (negative electrode) of a conventional lithium-ion cell is typically graphite made from carbon. The cathode (positive electrode) is typically a metal oxide. The electrolyte is typically a lithium salt in an organic solvent.

Lithium-ion (Li-ion) batteries are inarguably the most popular type of rechargeable battery for consumer electronics. They can be used for a variety of products from mobile phones to cars, and their qualities are superior compared to other rechargeable batteries. At Night Searcher we use highqualitylithium-ion (Li-ion) batteries for all but a few of our rechargeable flashlights, searchlights, head torches, and floodlights, as they allow us to provide the high-performance, durable products our customers are used to. Below we've listed the biggest advantages of lithium-ion batteries from the customers' point of view and delved into the science behind each characteristic



Fig. No. 8. Lithium-ion Battery

4.6 L298N Motor Driver-

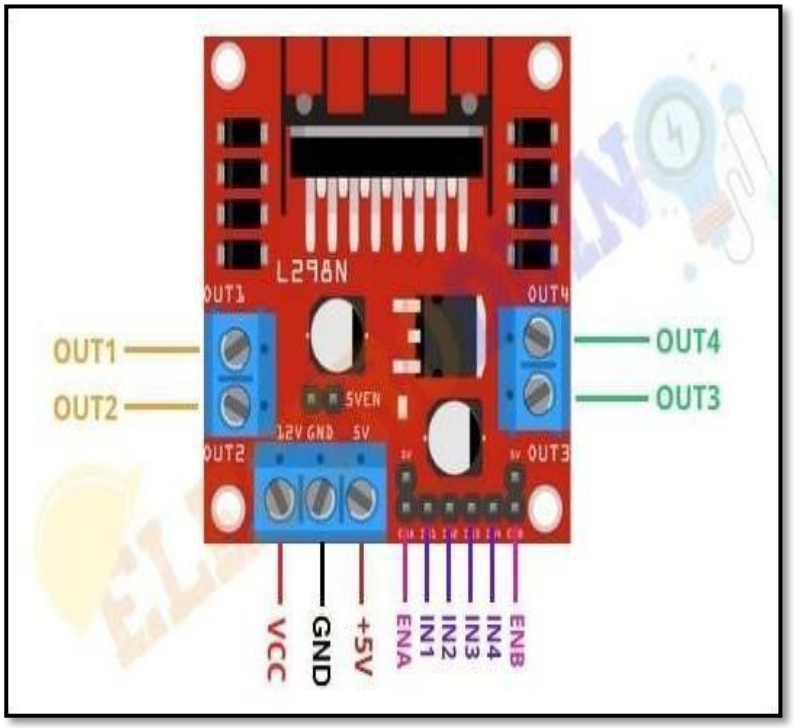


Fig. No. 9. L298N Motor

➢ Description

The L298N motor driver is based on the H-bridge configuration (an H-bridge is a simple circuit that lets us control a DC motor to go backward or forward.), which is useful in controlling the direction of rotation of a DC motor.

➢ Features

- Operating Supply voltage up to 46V.
- Low Saturation Voltage.
- Total DC Current up to 4A.
- Logical \"0\" Input Voltage up to1.5 V (High noise Immunity)
- Overtemperature Protection.

➢ Specifications

- Power supply - DC 5V-35V
- Peak Current - 2A
- Operating Current Range- 0-36mA
- Logic Voltage- 5V

➢ Interfacing of L298N Motor with DC Motor

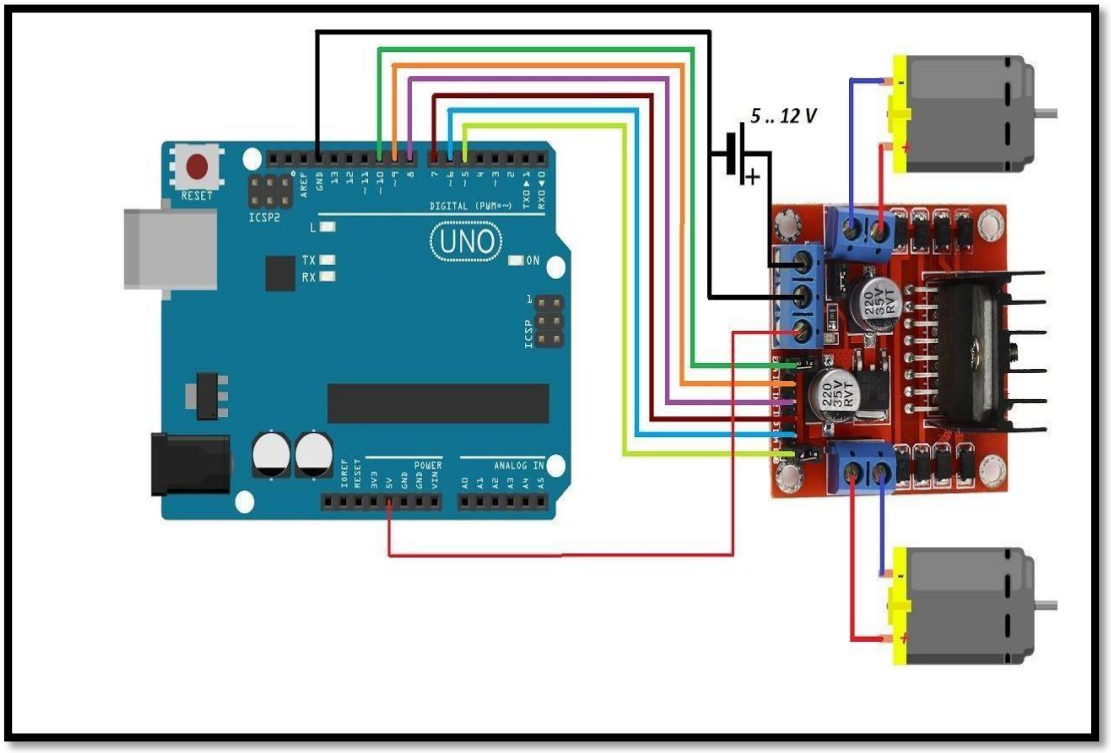


Fig. No. 10. Interfacing of L298N Motor

➢ L298N Module Pinout Configuration

IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B
ENA	Enables PWM signal for Motor A
ENB	Enables PWM signal for Motor B
OUT1 & OUT2	Output pins of Motor A
OUT3 & OUT4	Output pins of Motor B
12V	12V input from DC power Source
5V	Supplies power for the switching logic circuitry inside L298N IC
GND	Ground pin

4.7 Switch

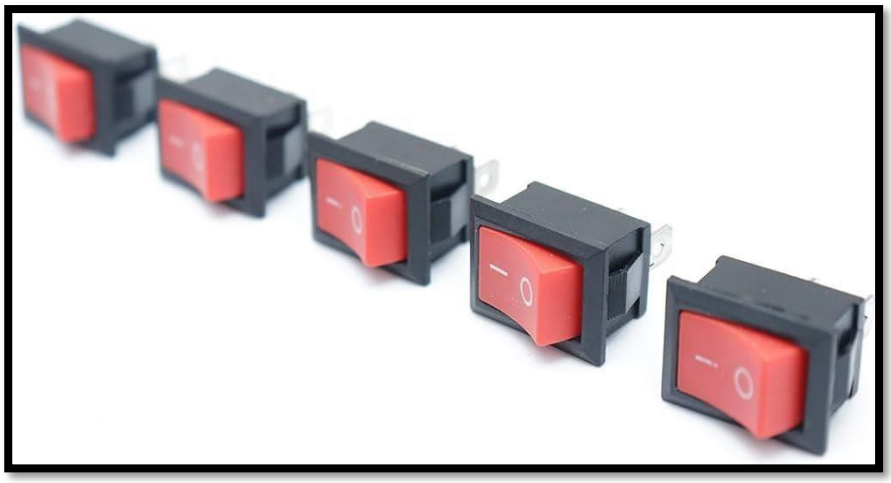


Fig. No.11. Switch

Switch is an electronic component that can disconnect or connect the conducting path in an electronic circuit, interrupting the electric current or diverting it from one conductor to another.

In electrical engineering, a switch is an electrical component that can disconnect or connect the conducting path in an electrical circuit, interrupting the electric current or diverting it from one conductor to another.[1][2] The most common type of switch is an electromechanical device consisting of one or more sets of movable electrical contacts connected to external circuits. When a pair of contacts is touching current can pass between them, while when the contacts are separated no current can flow.

Switches are made in many different configurations; they may have multiple sets of contacts controlled by the same knob or actuator, and the contacts may operate simultaneously, sequentially, or alternately. A switch may be operated manually, for example, a light switch or a keyboard button, or may function as a sensing element to sense the position of a machine part, liquid level, pressure, or temperature, such as a thermostat. Many specialized forms exist, such as the toggle switch, rotary switch, mercury switch, push-button switch, reversing switch, relay, and circuit breaker. A common use is control of lighting, where multiple switches may be wired into one circuit to allow convenient control of light fixtures. Switches in high-powered circuits must have special construction to prevent destructive arcing when they are opened.

4.8 Mecanum wheels

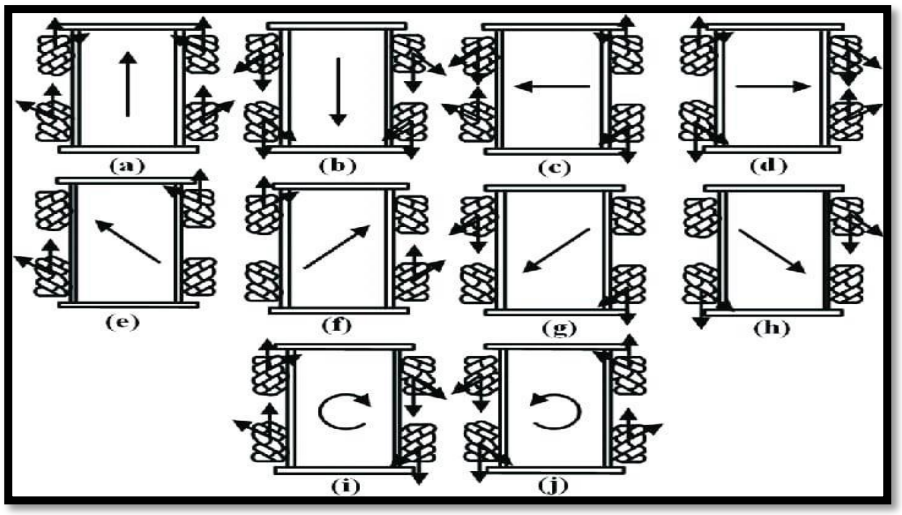


Fig. No.12. Mecanum wheel

Mecanum wheels are advance wheels used widely in the scope of robotics and often in competitive robotics due to their ability to introduce fully omnidirectional movement onto a drivetrain.

The Mecanum wheel feature a unique roller system that allow them to move in all directions, including diagonally and laterally.

This provides excellent maneuverability and easy moment.

The mecanum wheel is a form of tireless wheel, with a series of rubberized external rollers obliquely attached to the whole circumference of its rim. These rollers typically each have an axis of rotation at 45° to the wheel plane and at 45° to the axle line.[3] Each Mecanum wheel is an independent non-steering drive wheel with its own powertrain, and when spinning generates a propelling force perpendicular to the roller axle, which can be vectored into a longitudinal and a transverse component in relation to the vehicle.

4.9 3D Printed Robotic Arm



Fig. No. 13. Robotic Arm

A 3D printed robotic arm is a mechanical device consisting of joints, actuators, and links that are manufactured using 3D printing technology. It's typically controlled by a microcontroller or computer, allowing precise movement and manipulation of objects in various applications such as manufacturing, healthcare, and research. The advantage of 3D printing is the ability to customize and rapidly prototype different designs, making it cost-effective and adaptable for different

A robot arm for 3D printing works in a very similar way to a typical industrial robot arm, except that a printhead is fitted to the end of the arm. The robot arm has multiple joints, each one providing some freedom of movement, in total providing five- or six-axis control. The robot is then able to move, tilt, and position the printhead throughout a range of potential positions. In this way, the robot arm moves the printhead over a component in order to print multiple layers and contours.

5. Software Description

The Arduino Integrated Development Environment or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuine hardware to upload programs and communicate with them. Cut a small piece of PCB board according to the circuit.

➢ Writing Sketches:

Programs written using Arduino Software (IDE) are called **sketches.** These sketches are written in the text editor and are saved with the file extension. Ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right-hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

➢ Additional Commands:

Additional commands are found within the five menus: **File**, **Edit**, **Sketch**, **Tools**, **and Help**. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

➢ Sketch:

- Verify/Compile
- Checks your sketch for errors compiling it; it will report memory usage for code and variables in the console area.
- Upload
- Compiles and loads the binary file onto the configured board through the configured Port.
- Upload Using Programmer
- This will overwrite the boot loader on the board; you will need to use Tools > Burn Boot loader to restore it and be able to Upload to USB serial port again. However, it allows you to use the full capacity of the Flash memory for your sketch. Please note that this command will NOT burn the fuses. To do so Tools -> Burn Bootloader command must be executed.
- Export
- Compiled Binary Saves a .hex file that may be kept as archive or sent to the board using other tools.
- Show Sketch Folder
- Opens the current sketch folder.
- Include Library

Adds a library to your sketch by inserting #include statements at the start of your code. For more details, see libraries below. Additionally, from this menu item you can access the Library Manager and import new libraries from .zip files.

- Add File.
- Adds a source file to the sketch (it will be copied from its current location). The new file appears in a new tab in the sketch window. Files can be removed from the sketch using the tab menu accessible clicking on the small triangle icon below the serial monitor one on the right side of the toolbar.

➢ Tools:

- Auto Format**:** This format your code nicely: i.e., indents it so that opening andclosing curly braces line up, and that the statements inside curly braces are indented more.
- Archive Sketch: Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.
- Fix Encoding & Reload: Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
- Serial Monitor: Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.
- Board: Select the board that you're using. See below for description the various boards. Port. This menu contains all the serial devices (real or virtual) on your machine. Itshould automatically refresh every time you open the top-level tools menu.
- Programmer: For selecting a hardware programmer when programming a board orchip and not using the onboard USB-serial connection. Normally you won't need this, butif you're burning a boot loader to a new microcontroller, you will use this.
- Burn Bootloader: The items in this menu allow you to burn a boot loader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino orgenuine board but is useful if you purchase a new ATmega microcontroller (which normally comes without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the boot loader on the target board. This command also set the right fuses.

➢ Help:

- Here you find easy access to a number of documents that come with the Arduino Software (IDE). You have access to Getting Started, Reference, this guide to the IDE and other documents locally, without an internet connection. The documents are a local copy of the online ones and may link back to our online website.

➢ Sketchbook:

- The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the File > Sketchbook menu or from the Open button on the toolbar. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the Preferences dialog.

➢ Tabs, Multiple Files, and Compilation:

- Allows you to manage sketches with more than one file (each of which appears in its own tab). These can be normal Arduino code files (no visible extension), C files (.c extension), C++ files (.cpp), or header files (.h).

➢ Uploading:

- Before uploading your sketch, you need to select the correct items from the Tools>Board and Tools > Port menus. The boards are described below. On the Mac, the serial port is probably something like /dev/tty. usbmodem241 (for an Uno or Mega2560 or Leonardo) or /dev/tty.usbserial-1B1 (for a Demilune or earlier USB board), or /dev/tty. USA19QW1b1P1**.1** (for a serial board connected with a Key span USB-to-Serial adapter). On Windows, it is probably COM1 or COM2 (for a serial board) or COM4, COM5, COM7, or higher (for a USB board) - to find out, you look for USB serial device in the ports section of the Windows User.
- Device Manager. On Linux, it should be/dev/ttyACMx**,**/dev/ttyUSBx**or** similar. Once you have selected the correct serial port and board, press the upload button in the toolbar or select the Upload item from the File menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Decimal) that lack auto reset, you will need to press the reset button on the board just before starting the upload. On most boards, you will see the RX and TX LED blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

➢ Software Used to Control Module

- **MIT App Inventor**- MIT App Inventor (App Inventor or MIT AI2) is a high-level block-based visual programming language, originally built by Google and now maintained by the Massachusetts Institute of Technology. It allows newcomers to create computer applications for two operating systems: Android and iOS, which, as of 25 September 2023, is in beta testing. It is free and open-source released under dual licensing: a Creative Commons Attribution ShareAlike 3.0 Unported license and an Apache License 2.0 for the source code. Its target is primarily children and students studying computer programming, similar to scratch.

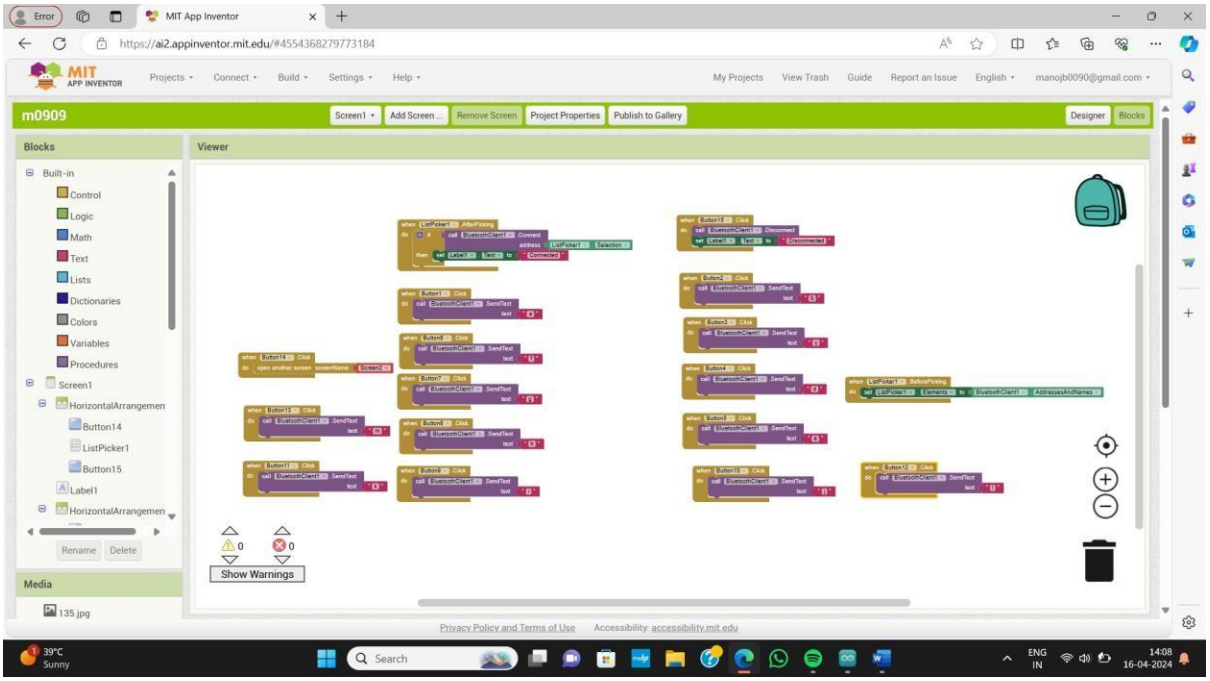


Fig.14. App used to control Pick & place Robotic Arm Vehicle

6. Program uploaded on Arduino Software

```
#include <SoftwareSerial.h>
#include <Servo.h>
Servo servo01;
Servo servo02;
Servo servo03;
Servo servo04;
Servo servo05;
Servo servo06;
SoftwareSerial Bluetooth(11, 12); // Arduino(RX, TX) - HC-05 Bluetooth (TX, RX)
int servo1Pos, servo2Pos, servo3Pos, servo4Pos, servo5Pos, servo6Pos; // current
position
int servo1PPos, servo2PPos, servo3PPos, servo4PPos, servo5PPos, servo6PPos; //
previous position
int servo01SP[50], servo02SP[50], servo03SP[50], servo04SP[50], servo05SP[50],
servo06SP[50]; // for storing positions/steps
int speedDelay = 20;
int index = 0;
int dataIn;
int m = 0;
void setup() {
pinMode(2, OUTPUT);
pinMode(3, OUTPUT);
pinMode(4, OUTPUT);
pinMode(5, OUTPUT);
pinMode(6, OUTPUT);
pinMode(7, OUTPUT);
pinMode(8, OUTPUT);
pinMode(9, OUTPUT);
  servo01.attach(A0);
  servo02.attach(A1);
  servo03.attach(A2);
  servo04.attach(A3);
  servo05.attach(A4);
  servo06.attach(A5);
  Bluetooth.begin(9600); // Default baud rate of the Bluetooth module
  Bluetooth.setTimeout(5);
  delay(20);
  Serial.begin(9600);
  // Move robot arm to initial position
  servo1PPos = 90;
  servo01.write(servo1PPos);
  servo2PPos = 60;
  servo02.write(servo2PPos);
  servo3PPos = 100;
  servo03.write(servo3PPos);
  servo4PPos = 120;
  servo04.write(servo4PPos);
```

```
  servo5PPos = 90;
  servo05.write(servo5PPos);
  servo6PPos = 140;
  servo06.write(servo6PPos);
}
void loop() {
  // Check for incoming data
  if (Bluetooth.available() > 0) {
    dataIn = Bluetooth.read(); // Read the data
   Serial.println(dataIn);
    if (dataIn == 16) {
      m = 16;
    }
    if (dataIn == 17) {
      m = 17;
    }
    if (dataIn == 18) {
      m = 18;
    }
    if (dataIn == 19) {
      m = 19;
    }
    if (dataIn == 20) {
      m = 20;
    }
    if (dataIn == 21) {
      m = 21;
    }
    if (dataIn == 22) {
      m = 22;
    }
    if (dataIn == 23) {
      m = 23;
    }
    if (dataIn == 24) {
      m = 24;
    }
    if (dataIn == 25) {
      m = 25;
    }
    if (dataIn == 26) {
      m = 26;
    }
    if (dataIn == 27) {
      m = 27;
    }
 if(dataIn == 'e') //stop
{ digitalWrite(2, LOW);
```

```
  digitalWrite(3, LOW);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(6, LOW);
  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
  digitalWrite(9, LOW);
    }
    if(dataIn== 'b') //forword
    {
  digitalWrite(2, HIGH);
  digitalWrite(3,  LOW);
  digitalWrite(4, HIGH);
  digitalWrite(5,  LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7,  LOW);
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
}
if (dataIn== 'd') //left

{
  digitalWrite(2,  LOW);
  digitalWrite(3,  LOW);
  digitalWrite(4, HIGH);
  digitalWrite(5,  LOW);
  digitalWrite(6,  LOW);
  digitalWrite(7,  LOW);
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
}
if (dataIn== 'h') //back
{

  digitalWrite(2,  LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4,  LOW);
  digitalWrite(5, HIGH);
  digitalWrite(6,  LOW);
  digitalWrite(7, HIGH);
  digitalWrite(8,  LOW);
  digitalWrite(9, HIGH);
}
if (dataIn== 'f') //right
{
  digitalWrite(2, HIGH);
  digitalWrite(3,  LOW);
  digitalWrite(4, LOW);
```

```
  digitalWrite(5,  LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7,  LOW);
  digitalWrite(8,  LOW);
  digitalWrite(9, LOW);
}
if (dataIn == 'j') //sl
{
   digitalWrite(2, LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, LOW);
  digitalWrite(6, HIGH);
  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
  digitalWrite(9, HIGH);
}
if (dataIn== 'k') //sr
{
   digitalWrite(2, HIGH);
  digitalWrite(3, LOW);
  digitalWrite(4, LOW);
  digitalWrite(5, HIGH);
  digitalWrite(6, LOW);
  digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
}
if (dataIn== 'l') //360r
{
  digitalWrite(2, HIGH);
  digitalWrite(3,  LOW);
  digitalWrite(4,  LOW);
  digitalWrite(5, HIGH);
  digitalWrite(6, HIGH);
  digitalWrite(7,  LOW);
  digitalWrite(8,  LOW);
  digitalWrite(9, HIGH);
}
if (dataIn== 'm') //360l
{
  digitalWrite(2,  LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5,  LOW);
  digitalWrite(6,  LOW);
  digitalWrite(7, HIGH);
  digitalWrite(8, HIGH);
```

```
  digitalWrite(9, LOW);
}
if (dataIn== 'a') //45
{ digitalWrite(2, LOW);
digitalWrite(3, LOW);
digitalWrite(4, HIGH);
digitalWrite(5, LOW);
digitalWrite(6, HIGH);
digitalWrite(7, LOW);
digitalWrite(8, LOW);
digitalWrite(9, LOW);}
if (dataIn== 'c') //135
{
  digitalWrite(2, HIGH);
  digitalWrite(3,  LOW);
  digitalWrite(4,  LOW);
  digitalWrite(5,  LOW);
  digitalWrite(6,  LOW);
  digitalWrite(7,  LOW);
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
}
if (dataIn== 'g') //315
{
  digitalWrite(2,  LOW);
  digitalWrite(3,  LOW);
  digitalWrite(4,  LOW);
  digitalWrite(5, HIGH);
  digitalWrite(6,  LOW);
  digitalWrite(7, HIGH);
  digitalWrite(8,  LOW);
  digitalWrite(9, LOW);
  }
if (dataIn== 'i') //225
{
  digitalWrite(2,  LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4,  LOW);
  digitalWrite(5,  LOW);
  digitalWrite(6,  LOW);
  digitalWrite(7,  LOW);
  digitalWrite(8,  LOW);
  digitalWrite(9, HIGH);
}
    }
    while (m == 16) {
      if (Bluetooth.available() > 0) {
        m = Bluetooth.read();
```

```
  }
  servo01.write(servo1PPos);
  servo1PPos++;
  delay(speedDelay);
 }
 // Move servo 1 in negative direction
 while (m == 17) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo01.write(servo1PPos);
  servo1PPos--;
  delay(speedDelay);
 }
 // Move servo 2
 while (m == 19) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo02.write(servo2PPos);
  servo2PPos++;
  delay(speedDelay);
 }
 while (m == 18) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo02.write(servo2PPos);
  servo2PPos--;
  delay(speedDelay);
 }
 // Move servo 3
 while (m == 20) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo03.write(servo3PPos);
  servo3PPos++;
  delay(speedDelay);
 }
 while (m == 21) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo03.write(servo3PPos);
  servo3PPos--;
  delay(speedDelay);
 }
```

```
// Move servo 4
while (m == 23) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo04.write(servo4PPos);
  servo4PPos++;
  delay(speedDelay);
}
while (m == 22) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo04.write(servo4PPos);
  servo4PPos--;
  delay(speedDelay);
}
// Move servo 5
while (m == 25) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo05.write(servo5PPos);
  servo5PPos++;
  delay(speedDelay);
}
while (m == 24) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo05.write(servo5PPos);
  servo5PPos--;
  delay(speedDelay);
}
// Move servo 6
while (m == 26) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo06.write(servo6PPos);
  servo6PPos++;
  delay(speedDelay);
}
while (m == 27) {
  if (Bluetooth.available() > 0) {
    m = Bluetooth.read();
  }
  servo06.write(servo6PPos);
```

```
      servo6PPos--;
      delay(speedDelay);
    }
    // If arm speed slider is changed
    if (dataIn > 101 & dataIn < 250) {
      speedDelay = dataIn / 5; // Change servo speed (delay time)
    }
}
```

7. List of Components

Sr. no	Components	Specification	Quantity
1	Arduino Nano	ATmega328p	1
2	Motor Driver	L298N	2
3	Servo Motor	SG 90, MG996R	5
4	775DC Motor	12V, 1500-35000rpm	4
5	Bluetooth Module	HC 05	1
6	Lithium-ion Battery	3.7V, 2600mAH	10
7	Switch	240V, 20A	1
8	Mecanum Wheel	360degree Rotation	4
9	Robotic Arm	3D Printed Arm	1

8. Advantages

- 4 Servo motor in Robotic Arm for easy pick and place.
- Android App control via Bluetooth.
- Mecanum wheel chassis for all direction movement.

9. Applications

- Pick and Place, Stacking and Sorting
- Product Assembly
- Part Inspection
- Packing

10. Future Scope

As far as the future scope of the project is concerned, we could replace the camera with an IR camera and an IR torch for enabling night vision.

- We also added some special material for the wheel strength.
- Due to its all-direction movement, it can be used in future automobile industries and also replace the normal tires.

We could also add some important sensor for better usage sensors like motion, thermal, ultrasonic etc.

- It can be used in future in space mission because it can work all direction and the robotic arm also help to pick and save the object.

11. Conclusion

The main advantage of this type of wheel is represented by the omni directional property that it provides, allowing extreme maneuverability and mobility in congested environments .In order to improve the wheel design is described. The maneuverability provided by omni directional vehicles can be utilized and can be very important both out door applications, such as search and rescue missions, military activities, planetary explorations and mine operations, long loads transportation, and indoor applications, like small good s transportation, powered robotic wheel chairs or shopping carts. The robotic arm purpose is solely focused on the storing capability that is for lifting heavy loads and placing it in the storage section in the industrial work environment. When coupled with omni directional mobility the storing capability and efficiency in industrial section may increase drastically. The main advantage in using this type of vehicle is the it requires a very small area for its turning of a full 360.

12. References-

- https://nevonprojects.com/remote-controlled-pick-place-robotic-vehicle/
- https://www.elprocus.com/pick-n-place-robot/
- https://www.elprocus.com/pick-n-place-robot/

13. Final Project-

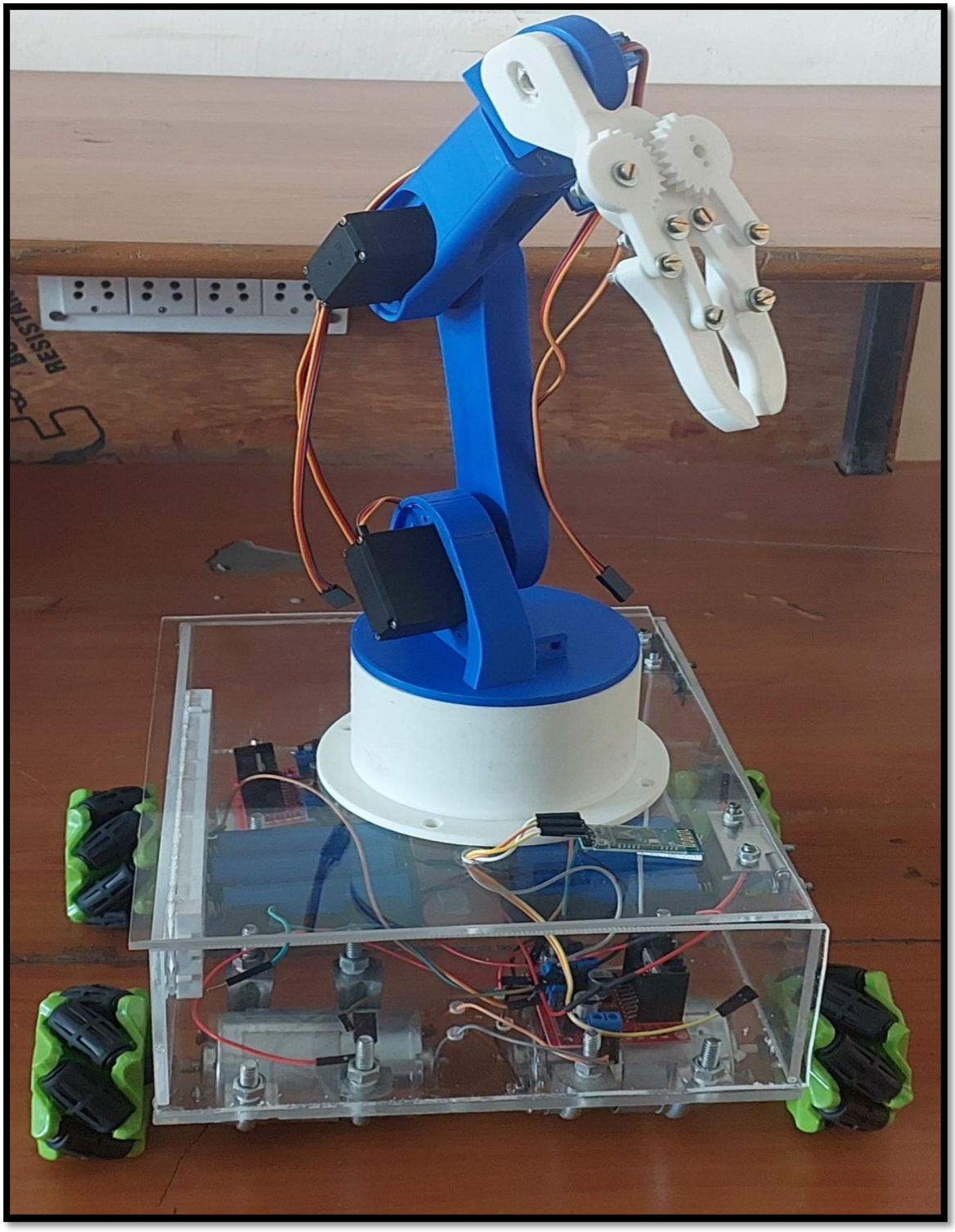


Fig.14. Pick and Place Robotic Arm

14. Group photo-



Image No.15 - From Left Mr. Atharva karatkar, Mr. Margur L. S. (Lecturer), Mr. Bagban S. R. (H.O.D of E&TC department), Mr. Ansh Jadhav, Mr. Shrinivas Kulkarni.